

Overparametrization for Landscape Design in Non-convex Optimization

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Question

- 1 Why is (stochastic) gradient descent (GD) successful? Or is it just “alchemy”?

- 1 Introduction
- 2 Saddlepoints and Gradient Descent
- 3 Landscape Design via Overparametrization
- 4 Generalization

(Sub)-Gradient Descent

Gradient Descent algorithm:

$$x_{k+1} = x_k - \alpha_k \partial f(x_k).$$

Non-smoothness

Deep Learning Loss Functions are not smooth! (e.g. ReLU, max-pooling, batch-norm)

Convergence of sub-gradient method to stationary points is only known for weakly-convex functions ($f(x) + \frac{\lambda}{2} \|x\|^2$ convex).

Theorem (Davis, Drusvyatskiy, Kakade, and Lee)

Let x_k be the iterates of the stochastic sub-gradient method. Assume that f is locally Lipschitz (and semialgebraic), then every limit point x^ is critical:*

$$0 \in \partial f(x^*).$$

- Difficulty is in the downward “kinks” like $(1 - \text{ReLU}(x))^2$
- Convergence rate is polynomial in $\frac{1}{\epsilon}, d$ to ϵ -subgradient.
- Clarke subgradient can be efficiently computed using Automatic Differentiation in $6x$ cost as function evaluation (Drusvyatskiy, Kakade and Lee 2018)

Theorem (Lee et al., COLT 2016)

Let $f : \mathbf{R}^n \rightarrow \mathbf{R}$ be a twice continuously differentiable function with the strict saddle property, then gradient descent with a random initialization converges to a local minimizer or negative infinity.

- Theorem applies for many optimization algorithms including coordinate descent, mirror descent, manifold gradient descent, and ADMM (Lee et al. 2017 and Hong et al. 2018)
- Stochastic optimization with injected isotropic noise finds local minimizers in polynomial time (Pemantle 1992; Ge et al. 2015, Jin et al. 2017)

Why are local minimizers interesting?

All local minimizers are global for the following problems:

- 1 ReLU networks via landscape design (GLM18)
- 2 Matrix Completion (GLM16)
- 3 Rank k approximation
- 4 Matrix Sensing (BNS16)
- 5 Phase Retrieval (SQW16)
- 6 Orthogonal Tensor Decomposition and Non-orthogonal Tensor Decomposition (GHJY15, Lee18)
- 7 Dictionary Learning (SQW15)
- 8 Max-cut via Burer Monteiro (BBV16, Montanari 16)
- 9 Overparametrized Deep Networks (DL18, LSL18)

Practical Landscape Design - Overparametrization

Over-parametrization

If back-propagation is not finding a low training error solution, then fit a bigger model.

Problem

How much over-parametrization do we need to efficiently optimize?

Over-parametrization Hypothesis

Optimization is “easy” when parameters $>$ sample size.

- Soudry and Carmon 2016 justified this for ReLU networks.
- Livni et al. empirically investigated this.
- Nguyen and Hein proved that when $k > n$, then all local are global.
- Liang et al. proved a similar result under a two subspace assumption on the p_X .

Why Quadratic Activation?

Case Study: Quadratic Activation Networks

$$f(x; W) = \sum_{i=1}^k \phi(w_i^T x),$$

where $\phi(z) = z^2$.

These can be formulated as matrix sensing with $\mathcal{X}_i = x_i x_i^T$.

Regularized Loss

$$\min_W \sum_i \ell(f(x_i; W), y_i) + \frac{\lambda}{2} \|W\|_F^2.$$

How much Over-parametrization?

- For $k \geq d$ that all local are global; relies on $y = x^T W^T W x = x^T M x$ for $M = W^T W$ (Haefele and Vidal, Bach, Burer-Monteiro)
- The result is independent of n , which is counter-intuitive. Can we get closer to $\# \text{ params} = kd > n$?

Random Regularization

$$L_C(W) = \sum_i \ell(f(x_i; W), y_i) + \frac{\lambda}{2} \|W\|_F^2 + \langle C, W^T W \rangle,$$

where C is random Gaussian $\mathcal{N}(0, \sigma^2)$.

Theorem

Let ℓ be a convex loss function, $\lambda > 0$, and $\sigma > 0$. If $k \geq \sqrt{2n}$, then almost surely all local min are global minima.

- Applies for arbitrarily small perturbation σ . By choosing σ small, we can closely approximate the solution of the unperturbed objective.
- Motivated by work on SDP (Burer & Monteiro, Boumal-Voroniski-Bandeira) which show that $k \geq \sqrt{2n}$ all non-degenerate local minima are global. Smoothing allows us to remove the non-degenerate local minima.
- Surprisingly, the same smoothing works even though our objective is not SDP-representable.

How about Generalization?

Generalization

The regularizer $\|W\|_F^2$ corresponds to $\|W^T W\|_*$. Small nuclear norm leads to generalization via standard Rademacher complexity bounds.

Corollary

Assume that $y = \sum_{i=1}^{k_0} \sigma(w_i^T x)$, and $x_i \sim \mathcal{N}(0, I)$. Then for $n \gtrsim \frac{dk_0^2}{\epsilon^2}$,

$$L_{te}(W) - L_{tr}(W) \leq \epsilon.$$

The sample complexity is independent of k , the number of neurons.

Quadratic Activation Network

- ① Training Error: Over-parametrization makes the optimization easy, since all local are global.
- ② Test Error: The generalization is not hurt by over-parametrization. The sample complexity only depends on k_0 , the number of effective neurons, and not k , the number of neurons in the model.

How do we show this for ReLU activations and deeper networks?

Optimization is only half the story

- Modern networks are over-parametrized meaning $p \gg n$ ($\frac{p}{n} \in (10, 200)$).
- Over-parametrization allows SGD to drive the training error to 0. But shouldn't the test error be huge due to overfitting?

Experiment

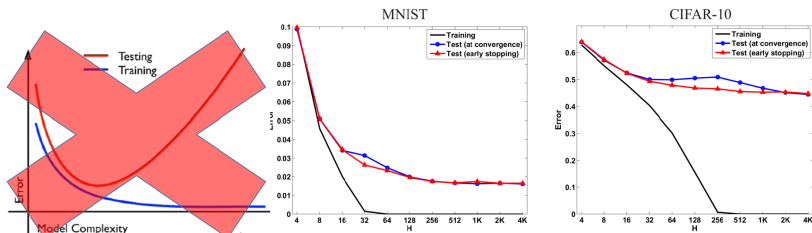


Figure: Credit: Neyshabur et al. See also Zhang et al.

- $p \gg n$, no regularization, no early stopping, and yet we do not overfit.
- Unclear what is the correct measure of model complexity. Clearly, parameter counting is not appropriate for SGD.

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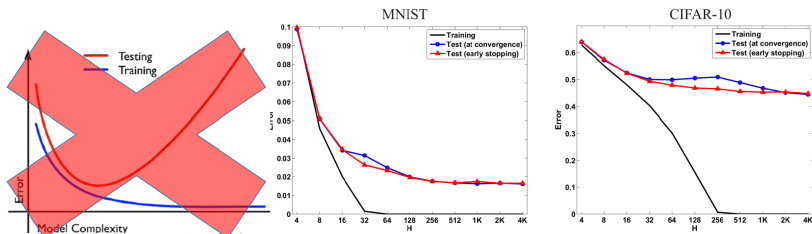


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- $p \gg n$, no regularization, no early stopping, and yet we do not overfit.
- Unclear what is the correct measure of model complexity. Clearly, parameter counting is not appropriate for SGD.
- Or is there regularization? Since $p \gg n$, there is a $p - n$ -dimensional space of global minima, and definitely some of these do not generalize.

Definition (Separable Data)

We will assume that $y_i(x_i^T w) > 0$ for some w .

- Equivalent of the over-parametrized regime in linear models. If $p \gg n$, this holds for almost all $\{x_i\}$.
- When the data is separable, there are infinitely many linear separators.

Implicit Regularization (via choice of Algorithm)

Warm-up: Logistic Regression with separable data

Gradient descent with any initial point w_0 on

$$\mathcal{L}(w) = \sum_i \log(1 + \exp(-y_i x_i^T w))$$

converges in direction to the ℓ_2 -SVM solution. In equations,

$$\frac{w(t)}{\|w(t)\|} \rightarrow C \arg \min_{y_i w^T x_i \geq 1} \|w\|_2.$$

(Soudry et al. 2018, Ji & Telgarsky 2018, Gunasekar et al. 2018)

This means that if the data is separable with a large margin, then GD+Logistic Regression generalizes as well as SVM.

Steepest Descent

$$w(t+1) = w(t) + \alpha \Delta w(t)$$

$$\Delta w(t) = \arg \min_{\|v\| \leq 1} v^T \nabla L(w(t)).$$

Coordinate descent is steepest descent wrt $\|\cdot\|_1$ and signed gradient method is steepest descent wrt $\|\cdot\|_\infty$.

Theorem (Gunasekar, Lee, Soudry, and Srebro)

On separable data, steepest descent converges in direction to the $\|\cdot\|$ -SVM solution, meaning $\frac{w(t)}{\|w(t)\|} \rightarrow C \arg \min_{y_i w^T x_i \geq 1} \|w\|$.

- Solution depends on the choice of algorithm.
- For coordinate descent, it is already known from the boosting literature that AdaBoost achieves the minimum ℓ_1 norm solution (Ratsch et al. 2004, Zhang and Yu 2005, Telgarsky 2013). Also related to the study of LARS algorithms.
- For ℓ_2 norm, this recovers the theorem before.

Steepest Descent in ℓ_2

Let us consider the ODE

$$\frac{dw(t)}{dt} = -\nabla \mathcal{L}(w(t)).$$

Steepest Descent maximally decreases the loss function per unit of arc-length as measured by

$$\|w(t)\| = \left\| \int_{s=0}^t -\Delta w(s) ds \right\| \leq \int_{s=0}^t \|\Delta w(s)\| ds.$$

Can upper bound margin by the loss

$$w^T x_i \geq -\log \mathcal{L}(w_t).$$

Thus by “spending” the least amount of ℓ_2 -norm, gradient descent maximally decreases the function.

Quadratic Activation Neural Net and Matrix Sensing/Completion

Setup

Given data (X_i, y_i) , and $\ell \in \{\exp, \log\}$,

$$\mathcal{L}(U, V) = \sum_i \ell(-y_i \langle X_i, UV^T \rangle).$$

We will focus on the case where U, V are fat, so over-parametrized.

- Multi-output one-hidden layer network with no non-linearity.
- One-hidden layer network with quadratic activation.
- One-bit Matrix sensing and completion.

At each step, we can re-parametrize with $W(t) = U(t)V(t)^T$. The equivalent problem is

$$\bar{\mathcal{L}}(W) = \sum_i \ell(-y_i \langle X_i, W \rangle).$$

These are equivalent optimization problems, but the dynamics are not equivalent. GD on $\mathcal{L}(W)$ and GD on $\bar{\mathcal{L}}(W)$ will give very different solutions.

Warm-up: Simple, but boring case

GD on $\bar{\mathcal{L}}$

What happens when you do GD on $\bar{\mathcal{L}}(W)$?

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GD on $\bar{\mathcal{L}}$

What happens when you do GD on $\bar{\mathcal{L}}(W)$?

Answer

Answer: This leads to an implicit $\|W\|_F$ regularizer, as seen by the previous section.

N.B. This does not correspond to what is done in practice. We care more about GD on $\mathcal{L}(U, V)$.

GD on $\mathcal{L}$

What happens you do GD on $\mathcal{L}(U, V)$?

GD on $\mathcal{L}$

What happens you do GD on $\mathcal{L}(U, V)$? Natural Guess: This corresponds to the regularizer $\|U\|_F^2 + \|V\|_F^2$, and

$$\min_{W=UV^T} \|U\|_F^2 + \|V\|_F^2 = \|W\|_{\text{tr}}.$$

Conjecture

When data is separable, gradient descent on U, V converges to the minimum nuclear norm solution, i.e.

$$\lim_t W(t) \stackrel{\text{dir}}{=} \arg \min_{y_i \langle X_i, W \rangle \geq 1} \|W\|_{\text{tr}}.$$

What we can prove

Theorem (Gunasekar, Lee, Soudry and Srebro 2018)

Gradient descent iterates $\lim_t U(t), V(t)$ are (up to scaling) first-order stationary points of $\arg \min_{\langle X, UV^T \rangle \geq \gamma} \|U\|_F^2 + \|V\|_F^2$

More generally,

Theorem (Gunasekar, Lee, Soudry and Srebro 2018)

For any homogeneous polynomial p , GD on

$$\sum_i \exp(-y_i \langle p(w), \mathcal{X} \rangle)$$

converges to a first-order stationary point of

$$\begin{aligned} \min & \|w\|_2 \\ \text{st } & \langle p(w), \mathcal{X} \rangle \geq 1 \end{aligned}$$

Implicit Regularization

- 1 Overparametrize to make training easy, but there are infinitely many possible global minimum
- 2 The choice of algorithm and parametrization determine the global minimum.
- 3 Generalization is possible in the over-parametrized regime with no regularization by choosing the right algorithm.
- 4 We understand only very simple problems and algorithms.

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- ① Gunasekar, Lee, Soudry and Srebro, *Characterizing Implicit Bias in Terms of Optimization Geometry*.
- ② Du and Lee, *On the Power of Over-parametrization in Neural Networks with Quadratic Activation*
- ③ Davis, Drusvyatskiy, Sham Kakade, and Jason D. Lee, *Stochastic subgradient method converges on tame functions*.
- ④ Lee, Panageas, Piliouras, Simchowitz, Jordan, and Recht, *First-order Methods Almost Always Avoid Saddle Points*.
- ⑤ Lee, Simchowitz, Jordan, and Recht, *Gradient Descent Converges to Minimizers*.